

Context

- Regardless of the importance of a language, many African languages remain under-represented and this fact holds for the siSwati language.
- With it being used for literally everything in Eswatini, from learning, verbal communication, writing, and everything else, exist limitation when it comes to Natural Language Processing (NLP) and Artificial Intelligence(AI).
- This hinders the potential benefits that these technologies offer.

Motivation

- Lately, we noticed the apparent revolution caused by AI and this does not leave behind languages and that motivated this study.
- With this study, we propose a siSwati dataset, cleaner, and tools that can be used to leverage si NLP
- Further, we lay a foundation for researchers, developers to work with and develop sentiment analysis in siSwati.

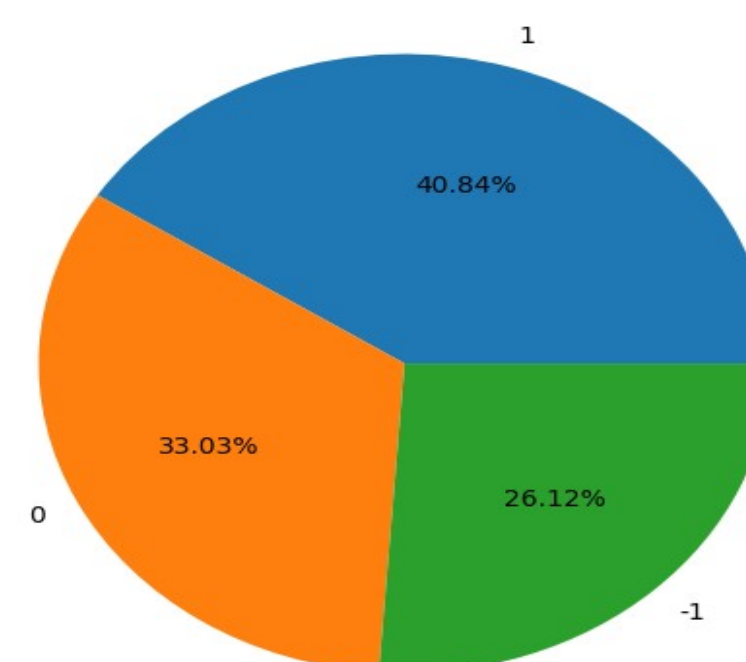
siSwati Language

- The language which is the solo, official language in Eswatini but also spoken in the Mpumalanga province of South Africa holds huge significance in as far as communication, culture, and tradition is concerned.
- It belongs to the Nguni subgroup of the Niger-Congo language family so it holds a lot of similarity with some southern African languages.
- siSwati is one of the thousands of languages housed in Africa which are regarded to be low-resourced or under-represented when looked at from the perspective of the availability of annotated datasets, models, researchers, and other tools mandatory for the accomplishment of NLP tasks.
- This hinders the use of Artificial Intelligence in strategic points of the economy including marketing, politics, social sciences and so one.
- siSwati is spoken by an estimate of 2.4 million people.

siSwatiSent Dataset

- The data was collected from the YouTube platform by scraping comments to videos in which people express themselves in siSwati.
- To curate a representative dataset, we collected the data across comedy, sports, news and other channels with vital needs to emaSwati.
- Some light preprocessing had to be done just so to maintain the expressions of the users. This was mainly done with the help of the library we created.
- Two native speakers annotated the data and the sentences were annotated as: **positive (1), negative (-1), or neutral (0)**
- The table and pie chart below shows the statistics and distribution of the dataset, respectively.

Characteristic	siSwatiSent
#Sentences	1895
#Words	11837
#Unique words	3474
#Positive Sentences	774
#Negative Sentences	495
#Neutral Sentences	626
Average Comment Length	6 Words



Experiments & Results

The results below show each model's performance on siSwatiSent dataset on the mention metrics.

Model	acc	bal. acc	recall	precision	f1-score
LSTM	40.4%	37.8%	40.4%	42.4%	35.5%
BI-LSTM	43.8%	38.8%	43.8%	37.7%	37.4%
SVM	55.2%	51.9%	55.2%	53.9%	53.6%
Naïve Bayes	53.8%	52.4%	53.8%	54.4%	54.0%
AdaBoost	49.0%	47.7%	49.0%	52.4%	42.2%
XGBoost	49.6%	46.9%	49.6%	49.7%	48.5%
mBERT	54.1%	52.6%	54.1%	56/2%	54.7%
SERENGETI LLM	61%	57.6%	61%	61.8%	61.3%

- Pre-trained multi-lingual models show better performance over all the ML and DL models.
- Deep Learning models require more data to improve their performance so that justifies why their performances are low.

Conclusion & Future Work

This work introduces a sentiment analysis dataset, and a siSwati specific library for cleaning text. Further to that, the dataset is publicly available for other researcher, analytical studies. In the future, we plan to expand the size of siSwatiSent, run experiments on other language models and also create a transformer model for the language which is capable for all other NLP tasks.